Dataset proposal & provenance

Chosen track: weather and environmental intelligence. Exact organizer filename: datasetpropsal.pdf.

What is being predicted?

The target is measured temperature in Celsius at the same station exactly 24 hours after an input observation. The three supported airport stations are OPKC (Karachi), OPLA (Lahore), and OPMT (Multan). This scope does not include rain, pollution, every neighborhood or all Pakistan.

Source and credible resource	Actual role / verification
NOAA NCEI Integrated Surface Database ncei.noaa.gov/products/land-based-station/integrated-surface-database	Observed airport weather and supervised target. Twelve station-year raw snapshots cover 2022-2025. Original filenames and hashes are in the warehouse manifest.
Open-Meteo historical snapshots open-meteo.com/en/docs/historical-weather-api	Three stored historical comparison datasets. Exact original request URL, selected model and original fetch time remain unverified. Do not describe them as proven JMA training inputs.
Open-Meteo previous-runs API open-meteo.com/en/docs/previous-runs-api	JMA GSM prior-day forecast fields used by the deployed frozen model. Reproduced Jan-Mar 2024 validation/calibration requests now have URLs, fetch times and response hashes. Original fitting feature table/search history is missing.
Aviation Weather Center API aviationweather.gov/data/api/	Current METAR station inputs for hourly inference only. Not training data; retrieval receipts record the live request.

Warehouse and split design

The repository contains 15 raw snapshots, 43,927 raw NOAA records (39,305 cleaned) and 78,912 Open-Meteo records. Final supervised splits contain 16,463 training, 1,783 validation, 755 calibration, 5,346 locked-test, 2,370 replay, 6,958 reserve and 124 boundary rows. Time ordering and 24-hour target boundaries avoid future-target leakage. Training tables shown to agents exclude held-out splits and targets.

Quality and reproducibility

A single make data command builds raw-reference, clean and processed outputs with manifest.csv and 14 warehouse checks. Mateen's unchanged agent separately removes duplicate rows, interpolates short interior gaps (up to three rows), flags suspect weather values and creates quality_log. Interpolation may use later rows and must not be introduced across training/test boundaries. Raw evidence is retained; source fingerprints verify the agent copies.

Feature and evaluation contract

The selected existing HGB artifact predicts a temperature correction using the rounded station observation, JMA prior-day weather fields and calendar/station features. It is frozen; this audit did not refit it. A separate early Ridge attempt and a later station-only alternative did not beat persistence and are retained as historical experiments.

The declared baseline repeats the unrounded current NOAA temperature. On the same 1,783 validation rows: model MAE 1.101946 C; baseline MAE 1.235165 C. Rounded deployment-input baseline MAE is 1.243915 C, reported separately. January worsens; February improves. This is reused development validation, not new untouched evidence.

March calibration supplies city residual quantiles: Karachi 183 rows / 2.224 C, Lahore 398 / 3.084 C, Multan 174 / 2.271 C. Nominal 90% ranges do not establish independent future coverage. The seven-day block bootstrap improvement interval is -0.042 to 0.271 C and includes zero.

Credibility, permissions and known gaps

Public availability is not a blanket commercial license. Credit NOAA, the Aviation Weather Center, Open-Meteo and its upstream forecast providers. Follow each source's current terms and redistribution conditions. Open-Meteo's free offering is for noncommercial use; a paid service using the previous-runs API needs the appropriate commercial plan. No commercial API cost has been confirmed.

Unresolved: original Open-Meteo collection receipts and original JMA training-feature/search records. Fresh archived-response receipts prove this reproduction, not what the original collector received. Recover these from the collector; do not reconstruct a false original timestamp. Archive revisions and forecast-grid versus airport differences can change results.

Evidence to inspect

HeatShift_Warehouse/manifest.csv and audit/OPEN_METEO_PROVENANCE.md; evidence/final_audit/primary_model/receipts.json, metrics.json and row_predictions.csv; automation/evaluate_frozen_primary.py; hourly run artifacts containing manifest.csv, raw responses, clean tables, SQLite and quality/insight logs.

Primary references

NOAA ISD documentation
NOAA ISD open-data registry
Open-Meteo previous-runs API
Open-Meteo plans and noncommercial-use limits
Aviation Weather Center API

Team contributions

Waleed Irfan: developed and improved the prediction model. **Syed Mateen Ali:** developed, managed and maintained both AI agents: the Data Quality Agent and the Insight Agent. **Aman Khan Lodhi:** handled data cleaning, data warehouse management and related data preparation.

Roles supplied by Waleed Irfan. AI assistance is recorded separately in the cumulative AI use log.